

Presynaptic nerve terminals release neurotransmitters via synaptic vesicle exocytosis. This action is mediated by the membrane-bound target-SNARE (t-SNARE) proteins syntaxin and SNAP-25, and the vesicle-associated SNARE (v-SNARE) protein synaptobrevin. Membrane fusion is initiated when the t-SNAREs and v-SNAREs form an α -helix bundle known as the core-*trans*-complex, achieved by zipping SNARE proteins together to form a more stable *cis*-complex (Figure 1).

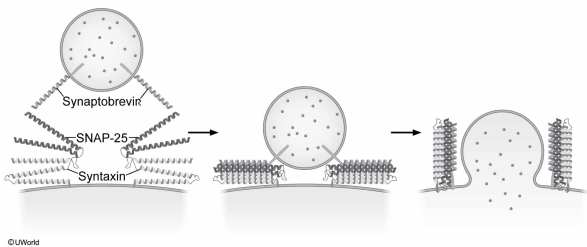


Figure 1 Formation of the core-*trans*-complex and subsequent membrane fusion

The *cis*-complex contains a region called the zero ionic layer, which is the main site of interaction between complexed proteins. Within the zero ionic layer, arginine on synaptobrevin coordinates with carbonyl groups present on the other residues shown in Table 1. The zero ionic layer is buried within amphipathic domains called leucine zippers, which shield the zero ionic layer from solvent molecules.

Table 1 Key Amino Acids of the *cis*-Complex Zero Ionic Layer

SNARE protein	Zero ionic layer residue
Synaptobrevin	R56
Syntaxin	Q226
SNAP-25 (Sn1)	Q53
SNAP-25 (Sn2)	Q174

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Once membrane fusion occurs, the *cis*-complex must disassemble. An early step in *cis*-complex disassembly is destabilization by the ATPase *N*-ethylmaleimide-sensitive factor (NSF), which disrupts leucine zipper regions, leading to vesicle reuptake. Movement of NSF to the cell membrane is mediated by the cytoplasmic protein α -SNAP, which recruits NSF after binding the N-terminal domain of syntaxin.

To determine the effect of synaptobrevin on the binding of α -SNAP to syntaxin, increasing amounts of recombinant His-tagged α -SNAP were added to a constant amount of syntaxin affixed to glutathione-agarose beads in the presence or absence of synaptobrevin. After incubation, bound α -SNAP was recovered and quantitatively analyzed by immunoblotting. Results are shown in Figure 2.

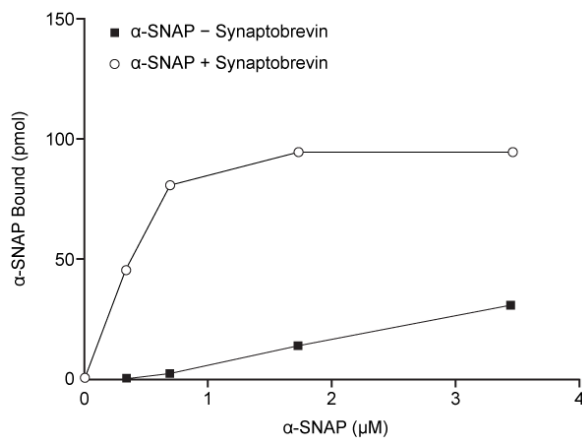


Figure 2 α-SNAP binding to syntaxin in the absence (square) or presence (circle) of synaptobrevin

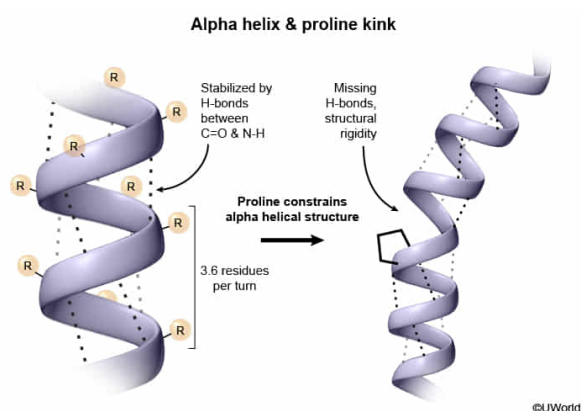
Which amino acid is LEAST likely to be found in normally functioning binding domains of t-SNARE and v-SNARE proteins?

- ☐ A. Asparagine [6%]
- ☐ B. Tryptophan [11%]
- ☐ C. Threonine [5%]
- ✓ ☒ D. Proline [76%]

Correct

76% answered correctly

Explanation:



As mentioned in the passage, the **interacting protein domains are α-helical**. Along with β-sheets, **α-helices** are common secondary structures found in proteins. α-Helices consist of single strands of protein arranged in spirals, with side groups extending outward. These helices include **3.6 amino acids per turn** and are stabilized by **hydrogen bonds** between carbonyl (C=O) and amide (N–H) groups at the n and $(n + 4)$ positions, respectively.

Proline and **glycine** are often found in loops and **linker regions** between α-helices and

β -sheets, particularly at **sharp turns** (eg, **beta turns**). Proline is unlikely to be found

within an α -helix because it is **structurally rigid** and introduces a kink in the chain that is useful for sharp turns but disrupts α -helices. In addition, the backbone amide of a proline residue connects back to its side chain to form a ring structure and is therefore unable to participate as a hydrogen bond donor. Glycine has the smallest side group of all the amino acids (a hydrogen atom) and is quite flexible. Although ideal for sharp turns, this flexibility causes α -helices to be too "floppy" and disrupts the structure.

(Choices A, B, and C) Both polar and nonpolar amino acids are found in α -helices. Polar amino acids such as **asparagine** and **threonine** are more likely to be found in regions exposed to water. Nonpolar amino acids such as **tryptophan** are more likely to be found within transmembrane regions, buried within a folded protein, or at sites of protein-protein interaction. Proline and glycine are the only amino acids that are not commonly found in α -helices.

Educational objective:

α -Helices have 3.6 amino acids per turn and are stabilized by hydrogen bonds between carbonyl and amide groups. Of the amino acids, proline is least likely to be found within α -helices and is more often found in linker regions or at sharp turns due to its structural rigidity. Glycine also tends to disrupt helices due to its excessive flexibility.

Time spent:0 sec

Subject:
Biochemistry

QID:400493

System:
Amino Acids and Proteins